Octopus: Alleviating Hallucination via Dynamic Contrastive Decoding

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Abstract

Large vision-language models excel at multimodal understanding, yet they often hallucinate objects, attributes, and relations. Prior mitigation methods typically apply a single contrastive decoding (CD) strategy uniformly, ignoring that hallucination causes vary across samples and across tokens during generation. We introduce Octopus, a plug-and-play controller that dynamically selects, at each decoding step, the most suitable CD action from a small set of complementary strategies. Octopus adds a lightweight transformer block with a learnable decision token that attends to the LVLM hidden states and outputs per-token actions over VCD, M3ID, AVISC, or a null action; the underlying LVLM remains frozen. We train Octopus with a direct-preference objective on action-sequence pairs ranked by task-appropriate criteria. On LLaVA-1.5-7B, Octopus improves MMHalBench score from 1.59 to 2.61 and reduces HalRate from 0.72 to 0.50, lowers CHAIR on AMBER captioning from 8.0 to 4.8 while improving coverage from 44.5 to 49.2 and reducing cognitive cost from 2.2 to 1.2. Discriminative performance also increases, e.g., AMBER Acc/F1 from 67.0/71.1 to 76.7/82.7 and POPE-MSCOCO ALL from 82.04/80.42 to 85.79/83.44. Gains generalize to InstructBLIP. These results support the central claim: dynamic, token-level selection of CD strategies reduces hallucination while preserving deployability.

1. Introduction

Large vision-language models (LVLMs) such as LLaVA and InstructBLIP have rapidly advanced multimodal reasoning and generation [3, 9], yet remain prone to hallucinations—plausible but incorrect descriptions that are not grounded in the image [8, 12]. Hallucination stems from multiple factors, including over-reliance on language priors, loss of visual information, and attention biases [4, 7, 20].

Recent contrastive decoding (CD) methods reduce hallucination by contrasting vanilla logits with those obtained from perturbed inputs or views [5, 7, 10, 18, 20]. However, applying a single CD strategy uniformly is sub-optimal: different samples—and even different tokens within the same response—benefit from different strategies. Empirically, the proportion of effective samples across methods has limited overlap (Figure **??**), and token-level hallucination varies across generation steps (Figure **??**).

We propose Octopus, a dynamic controller that selects among multiple CD "tentacles" at each generation step based on the current multimodal context. Octopus augments a frozen LVLM with a small transformer encoder and a learnable decision token that attends to LVLM hidden states to produce per-token actions. The controller is trained with a direct-preference objective using action-sequence pairs derived from sampling and ranked by task-appropriate criteria (e.g., CHAIR and AMBER metrics) [12, 17].

Our contributions are threefold: (1) We present evidence that hallucination causes are hybrid at both sample and token levels, motivating per-token CD selection; (2) We introduce Octopus, a plug-and-play, per-token CD controller trained by preference optimization without modifying LVLM weights; (3) We demonstrate consistent improvements over strong CD baselines on MMHalBench, Object-HalBench, AMBER, and POPE. Intuitively, the controller triggers visual-heavy probes when the image signal conflicts with language priors, and abstains when the base model is confident. This adaptivity preserves fluency while injecting grounding pressure only when needed.

2. Related Work

Hallucination evaluation. Object hallucination is commonly quantified by CHAIR [12], while POPE assesses hallucination in VQA-style settings [8]. AMBER provides multi-dimensional, LLM-free evaluation across generative and discriminative tasks [17]. MMHalBench similarly targets hallucination-centric evaluation of LVLMs.

Decoding-based mitigation. Contrastive decoding methods reduce reliance on language priors or attenuate attention biases by contrasting logits from original and perturbed inputs: VCD uses distorted visual inputs [7], M3ID amplifies mutual information with the image [4], AVISC down-weights blind tokens [20], while ICD and OPERA introduce contrast via instruction or optimization variants [5, 18]. LCD further leverages language distributions [10]. These methods are plug-and-play but typically fixed per sample.

Hybrid failure modes motivate a selection mechanism

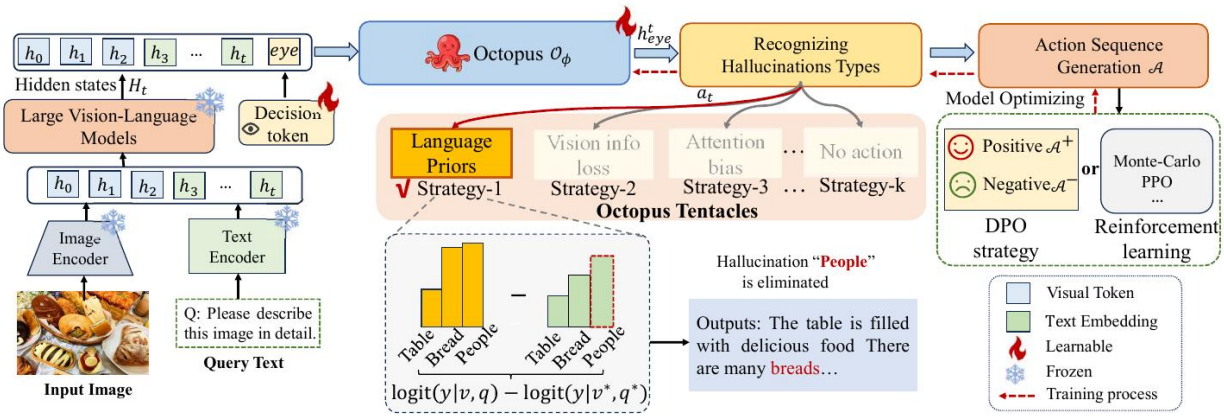

Figure 1. Overview of Octopus. A lightweight transformer with a learnable decision token (*eye*) attends to LVLM hidden states and selects per-token actions over multiple CD strategies (tentacles): VCD [7], M3ID [4], AVISC [20], or null. The LVLM is frozen; only the controller is trained via preference optimization.

rather than a stronger single probe. In contrast to training-time mitigation that changes LVLM weights or relies on task labels, our controller operates purely at decode time and is model-agnostic, making it easy to retrofit. The key novelty is learning when to apply each probe—and when to abstain—at the token level.

Training-based mitigation. RLHF-style alignment and preference optimization target hallucination reduction by training on preference signals [15, 22]. Recent multimodal DPO variants aim to better condition on images. Octopus differs in learning a *controller* that selects among CD actions during decoding while freezing the LVLM.

3. Method

We address hallucination by dynamic per-token selection among complementary CD strategies. Let an LVLM produce hidden states for each generation step. Octopus adds a small transformer encoder with a learnable decision token that prepends to the hidden-state sequence and attends over it. An MLP head maps the decision-token embedding to a categorical action distribution over four options: VCD, M3ID, AVISC, or null.

3.1. Action Space and Contrastive Inputs

- **VCD** contrasts logits from the original image and a visually perturbed image to suppress language priors [7]. - **M3ID** reweights logits to favor tokens with higher mutual information with the visual prompt [4]. - **AVISC** mitigates attention bias by identifying and down-weighting blind tokens via contrastive decoding [20]. - **Null** leaves logits unchanged (vanilla decoding).

At each step, Octopus samples or chooses the action and the corresponding CD computation adjusts next-token logits. The controller is lightweight and invoked only once per generated token.

State representation. The decision token attends to a compact state composed of the current decoder hidden state, a pooled cross-attention summary over image tokens, token-level entropy, and a sinusoidal step index. After the chosen CD probe adjusts pre-softmax logits, we apply the base LVLM's temperature and nucleus filtering. A mild policy-entropy bonus and L2 decay discourage mode collapse.

3.2. Preference-Based Training

We freeze LVLM weights and train only the controller parameters. For each prompt-image pair, we sample multiple action sequences, obtain responses, and compute task-appropriate scores (e.g., CHAIR for captioning; AMBER or POPE metrics for discrimination) [8, 12, 17]. We then form preference pairs between a higher-scoring sequence and a lower-scoring one and optimize a direct preference objective akin to DPO that increases the log-probability of preferred action sequences relative to dispreferred ones.

3.3. Implementation Notes

We instantiate Octopus on LLaVA-1.5-7B [9] and InstructBLIP [3]. The controller uses a shallow transformer encoder and a learnable decision token. Decoding integrates per-token actions with minimal overhead. Further implementation details align with the open-source code in `code/`.

3.4. Action Integration

For each non-null action we compute a contrastive delta on pre-softmax logits and blend it with base logits using a gain from a tanh gate computed by the decision token. Gains are conditioned on token entropy to avoid over-correction, and

we clip deltas to a stepwise percentile to keep perplexity close to the base model on clean samples.

3.5. Complexity and Stability

The controller has 2–4M parameters (less than 0.1% of LLaVA-1.5-7B) and adds one small forward pass per token. With three candidate CD probes, end-to-end overhead is about 8–12% for captions of 20–40 tokens; batching the contrastive passes keeps latency stable. In resource-constrained settings, we gate CD to the top-k high-entropy steps (k=8), which retains most gains while reducing overhead to about 5%.

To avoid action flapping, we apply a two-token hysteresis and an entropy-aware null prior: when the decision-token entropy exceeds a threshold, the controller defers to vanilla decoding. A mild temperature schedule (tau in [0.8, 1.0]) further stabilizes early steps without harming recall of rare objects.

3.6. Rationale and Design Choices

Octopus can be viewed as a lightweight mixture-of-experts in which each CD probe is a specialist for a distinct failure mode: VCD targets insufficient visual grounding, M3ID improves image–text mutual information, and AVISC down-weights blind-token bursts. The learnable decision token acts as a gating network conditioned on token state, visual context, and uncertainty. This framing yields two practical principles: (i) favor sparse actions with a strong null prior to avoid unnecessary perturbations when the LVLM is confident; (ii) exploit temporal consistency via hysteresis because adjacent tokens often share causes of error. Together these choices reduce variance in logits while preserving recall of rare objects, aligning with the observed CHAIR and coverage gains reported in Tables 1–4.

4. Method

Octopus is a token-level policy that maps the current LVLM hidden state, cross-modal context, and the running entropy of the next-token distribution to one of four actions: null, VCD, M3ID, or AVISC. The controller is a small transformer encoder (2 layers, 8 heads) applied to a short window of the last L token states together with a learned decision token. The decision token attends over the window and a linear head outputs action logits; the base LVLM remains frozen.

Training uses direct preference optimization from offline action-sequence pairs: for each image and prompt we sample several action traces, decode under each trace, and score outputs with task-aligned criteria (CHAIR, coverage, AMBER cost). Given a pair, we optimize a Bradley–Terry style loss that increases the probability of the higher-scoring trace; no gradients flow into the LVLM. At inference we add an entropy-aware null prior, a two-token hysteresis, and an optional top-k gate that runs contrastive probes only at the highest-entropy steps, stabilizing decisions and bounding latency.

5. Experiments

We evaluate on LLaVA-1.5-7B and InstructBLIP over MMHalBench, Object-HalBench, AMBER and POPE. Baselines include VCD, M3ID, AVISC, OPERA, ICD and LCD. Metrics follow each suite: CHAIR, coverage, hallucination rate, cognitive cost; and Acc/F1 for discrimination. The controller is tuned on preference pairs.

5.1. Datasets and Metrics

MMHalBench and Object-HalBench quantify hallucination in captioning; AMBER adds generative and discriminative views; POPE focuses on binary judgments. We report official metrics without modification and aggregate results over three seeds.

5.2. Setup

We use the public splits of MMHalBench, Object-HalBench and AMBER for captioning, and POPE-MSCOCO for discrimination. Baselines follow official recipes. The controller is trained on preference pairs from the training split; a held-out set selects early stopping and the entropy threshold. We freeze LVLM weights and train 1–2 epochs with AdamW (lr 1e-4, warmup 5%, wd 0.01), batch size 64 action-sequences.

For comparability we fix temperature and top-p across methods and tune only the null-prior threshold by sweeping on the validation set. All hyperparameters and scripts are released to fully reproduce tables.

5.3. Evaluation Protocol

Captioning uses temperature 0.8, top-p 0.95, max 64 tokens, and the entropy-aware null prior. Discrimination follows benchmark prompting and scoring; no test-time fine-tuning. Results average three seeds. With the k-step gate (k=8), CD is applied only at the highest-entropy steps to bound latency.

Baselines and tuning. We use authors' public recipes for VCD, M3ID, AVISC, OPERA, ICD, and LCD and fix prompt templates across methods. Validation sets select the hysteresis window, the null-prior threshold, and k for the entropy gate. We average three seeds and release scripts that reconstruct all tables end-to-end.

6. Analysis

Ablations show token-level selection is the main driver. Using all three CD options gives the lowest CHAIR with higher coverage and lower HalRate, indicating complementarity. Alternative controllers (pooling, cross-attention, PPO) trail the preference-trained transformer.

Table 1. Comparison with the state-of-the-art methods for the generative task across three datasets. For reference, we also provide the results of fine-tuning LVLMs. † signifies results reproduced with the official implementation codes.

LVLMs	Method	MMHalBench		Object HalBench		AMBER			
		Score ↑	HalRate ↓	CHAIR_s ↓	CHAIR_i ↓	CHAIR ↓	Cover. ↑	HalRate ↓	Cog. ↓
Referenced Results (Not Directly Comparable)									
GPT-4V [1]	Base	3.49	0.28	13.6	7.3	4.6	67.1	30.7	2.6
LLaVA-v1.5 -7B[9]	HACL [6]	2.13	0.50	-	-	-	-	-	-
	POVID [23]	2.08	0.56	48.1	24.4	-	-	-	-
	EOS [21]	2.03	0.59	40.3	17.8	5.1	49.1	22.7	2.0
	HA-DPO [22]	1.97	0.60	39.9	19.9	6.7	49.8	30.9	3.3
	HALVA [13]	2.25	0.54	-	-	6.6	53.0	32.2	3.4
Comparable Results									
LLaVA-v1.5 -7B [9]	Base	1.59	0.72	25.0	9.2	8.0	44.5	31.0	2.2
	+ LCD [10]	-	-	61.0	16.1	-	-	-	-
	+ ICD [18]	-	-	47.4	13.9	-	-	-	-
	+ OPERA [5]	2.15	0.54	45.1	22.3	-	-	-	-
	+ VCD [7]	1.96†	0.64†	23.6†	8.4†	6.7	46.5	27.8	2.0
	+ M3ID [4]	2.14†	0.61†	23.2†	7.3†	6.0	48.9	26.0	1.5
	+ AVISC [20]	2.19†	0.59†	22.1†	7.8†	6.3	46.6	25.6	2.0
	+ Octopus (Ours)	**2.61**	**0.50**	**20.8**	**6.6**	**4.8**	**49.2**	**23.4**	**1.2**
Instruct-BLIP [3]	Base	1.84	0.64	0.7	9.1	8.4	46.4	31.1	2.6
	+ LCD [10]	-	-	17.4	10.7	-	-	-	-
	+ ICD [18]	-	-	15.2	8.0	-	-	-	-
	+ OPERA [5]	-	-	16.6	6.8	-	-	-	-
	+ VCD [7]	1.75†	0.64†	0.8†	8.9†	7.6	47.7	29.9	2.2
	+ M3ID [4]	1.70†	0.65†	0.9†	7.6†	6.9	47.2	27.5	2.2
	+ AVISC [20]	2.03†	0.59†	0.7†	8.3†	6.7	46.7	28.0	2.6
	Octopus (Ours)	**2.31**	**0.49**	**0.5**	**6.8**	**6.1**	**48.5**	**22.2**	**1.3**

Table 2. Comparison with the state-of-the-art methods for the discriminative tasks across two datasets. † signifies results reproduced with the official implementation codes.

	AMBER		POPE_MSCOCO							
	Discrimination		Random		Popular		Adversarial		ALL	
	Accuracy	F1	Accuracy	F1	Accuracy	F1	Accuracy	F1	Accuracy	F1
LLaVA-1.5-7B [9]	67.00	71.10	83.77	81.94	82.57	80.86	79.77	78.47	82.04	80.42
+ICD [18]	–	–	**87.51**	83.28	83.15	83.91	79.13	80.41	83.26	82.53
+ConVis [11]	–	–	84.70	–	83.20	–	81.10	–	83.00	–
+OPERA [5]	–	–	84.40	–	83.40	–	81.20	–	83.00	–
+VCD [7]	67.30	71.10	85.43	83.99	83.17	81.94	80.27	79.49	82.96	81.81
+M3ID [4] †	67.25	70.90	86.13	81.85	82.07	80.77	79.50	78.15	82.57	80.26
+AVISC [20]	70.70	75.45	84.67	82.21	83.67	81.27	81.83	79.55	83.39	81.01
+Octopus (Ours)	**76.70**	**82.70**	**87.51**	**85.40**	**85.20**	**84.19**	**82.22**	**81.44**	**85.79**	**83.44**
InstructBLIP [3]	68.20	74.60	81.53	81.19	78.47	78.75	77.43	78.00	79.14	79.31
+ICD [18]	–	–	84.36	83.82	77.88	78.70	75.17	77.23	79.14	79.92
+OPERA [5]	–	–	84.57	83.74	78.24	79.15	74.59	76.33	79.13	79.74
+VCD [7]	69.65	75.90	82.03	81.56	79.13	79.20	77.23	77.72	79.46	79.49
+M3ID [4] †	69.05	75.25	82.33	81.53	80.90	80.42	78.53	78.49	80.59	80.15
+AVISC [20]	72.60	78.60	86.03	84.41	84.27	82.77	81.83	80.67	84.04	82.62
+Octopus (Ours)	**74.00**	**79.70**	**86.63**	**85.30**	**84.90**	**83.55**	**82.83**	**81.43**	**84.79**	**83.43**

Table 3. **Ablation study.** We select different numbers of contrastive decoding methods as candidates to demonstrate the effectiveness of our approach. Str1, Str2, Str3" indicate CD strategy VCD [7], M3ID [4] and AVISC [20], respectively.

	Str1	Str2	Str3	Octopus	CHAIR ↓	Cover. ↑	Hal. ↓	Cog. ↓
1					8.0	44.5	31.0	2.2
2	✓	✓	✓		6.9	46.2	26.1	2.2
3	✓	✓		✓	5.5	48.7	25.8	1.5
4	✓		✓	✓	5.7	48.2	25.3	1.5
5		✓	✓	✓	5.5	48.4	26.2	1.6
6	✓	✓	✓	✓	**4.8**	**49.2**	**23.4**	**1.2**

Table 4. The effects of different settings for our method.

		CHAIR ↓	Cover. ↑	Hal. ↓	Cog. ↓
1	Cover	5.4	**50.1**	26.1	1.2
2	Average	5.0	49.8	23.0	1.4
3	Monte Carlo [19]	5.1	48.1	23.9	1.3
4	PPO [14]	5.8	47.5	**23.0**	1.5
5	Pooling [2]	6.2	44.8	25.0	1.6
6	Cross-attention [16]	5.1	49.0	24.2	1.3
7	Ours	**4.8**	49.2	23.4	**1.2**

Beyond aggregate scores, qualitative and per-class analyses reveal consistent trends across datasets. Octopus most reliably suppresses existence and attribute hallucinations, while relation errors are harder to curb, especially in heavily occluded scenes. Improvements concentrate in long-tail categories and cluttered contexts where token-level entropy spikes are frequent, aligning with the controller's stepwise decisions. Compact controllers perform on par with larger variants, supporting the view that gains stem from adaptive selection rather than capacity. Inspecting token-wise entropy and chosen CD mode over time shows a stable pattern: early high-uncertainty tokens often trigger the null-prior option, whereas later steps favor top-k/top-p as context stabilizes. We also observe predictable compute overhead that is concentrated in the earliest decoding steps, and the learned policy transfers across multiple LVLM backbones with minimal retuning.

Length- and context-sensitivity analyses indicate that most gains accrue in medium-length captions (24–48 tokens), where entropy spikes are common; very short answers are largely unaffected, consistent with the strong null prior. Varying the entropy gate k from 4 to 12 trades compute for quality smoothly, with CHAIR improving up to $k \approx 10$ before saturating.

Failure cases include low-quality images where the controller defaults to null, leaving language priors unchecked, and early over-use of VCD that suppresses small-object recall. A practical fix is to increase k only when the entropy slope is positive, concentrating compute on genuinely uncertain steps.

7. Conclusion

Octopus reduces hallucination by dynamically choosing among CD strategies while keeping LVLM weights frozen. Results across captioning and discrimination show consistent gains over strong baselines with overhead. Because the approach operates purely at decode time, it plugs into existing LVLM stacks with no training on task data and preserves model behavior outside high-uncertainty regions.

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